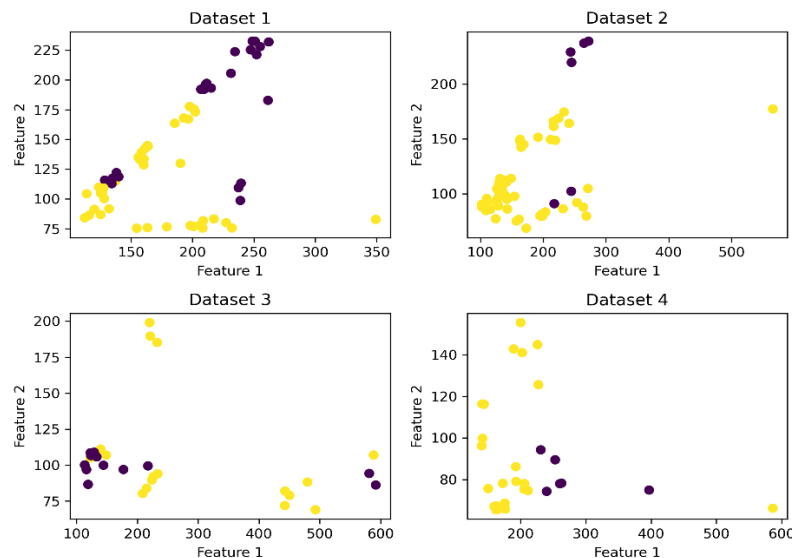


Machine Learning and Neural Computing:

Data Classification Coursework

1. Task 1: Data Exploration

- (a) I used `read_excel()` function to load all the Excel files (.xlsx) and convert them into DataFrame objects. The default parameters were used, including `header=0`, `sheet_name=0`, `index_col=None`. After loading the datasets, the feature columns and label column were separated using the `iloc` function. The expression `high_HNR_train.iloc[:,2:]` selects all rows and the 22 columns as features, while `high_HNR_train.iloc[:,1]` selects the second column as the label.
- (b) Scatter plot for two features



Plot description: In dataset 1, there seems to be a clear upward trend, showing a positive relationship between Feature 1 and Feature 2. Despite some overlap, there is also a clear separation between classes which may help in distinguishing the classes. In dataset 2, there also exists a positive correlation between both features. However, the greater overlap between the classes shows less clear separation between the classes. Also, an instance of outlier can also be seen in the graph. Moreover, Dataset 3 displays the data points being spread out more vertically showing higher variation in feature 2 values. Lastly, Dataset 4 shows two main clusters. One cluster shows moderate values of feature 1 and feature 2 and the other cluster shows higher feature 2 values. Some data points appear far from both clusters, indicating possible outliers.

Key Findings: Feature 1 and Features 2 show varying levels of correlation across datasets. Some datasets like Dataset 1 show clear class grouping while other datasets such as Dataset 2 exhibit greater overlap. Apart from this, the scatter plot clearly shows that the two features provide partial to not perfect separation of classes.

- (c) For the normalization of the test sets (High HNR test, Low HNR test), I have used StandardScaler from the sklearn.preprocessing. This transforms the mean of the feature close to zero and standard deviation of the feature close to 1.

The scaler was fitted on the training data. However, I used the same scaler to transform the test data. This ensures consistent scaling and avoid data leakage. Furthermore, I have calculated mean and standard deviation for both sets of test data. The mean is close to 0 while the standard deviation is close to 1. Below is the table of the output that shows both mean and standard deviation for test sets.

Metric	High-HNR Group	Low-HNR Group
Mean	-0.43861594149872646	0.31067677053217074
Standard Deviation	0.8898900935581869	0.5753608581399678

- (d) The table below demonstrates the percentage of variance explained by each of the first seven principal components for both PCAs.

Principal Component	High-HNR Variance (%)	Low-HNR Variance (%)
PC1	57.964195	57.811175
PC2	14.207906	14.928313
PC3	7.917365	7.815793
PC4	5.583167	6.843312
PC5	3.629992	3.479762
PC6	3.159114	2.579099
PC7	2.246254	2.039917

The table summarizes two different PCAs performed on two sets of training data (High HNR training set, Low HNR training set). The first principal component explains the largest proportion (57.96%) of variance in both datasets, followed by the second principal component (14.20%). Together, the first few components capture the majority of the variability in the data. Apart from this,

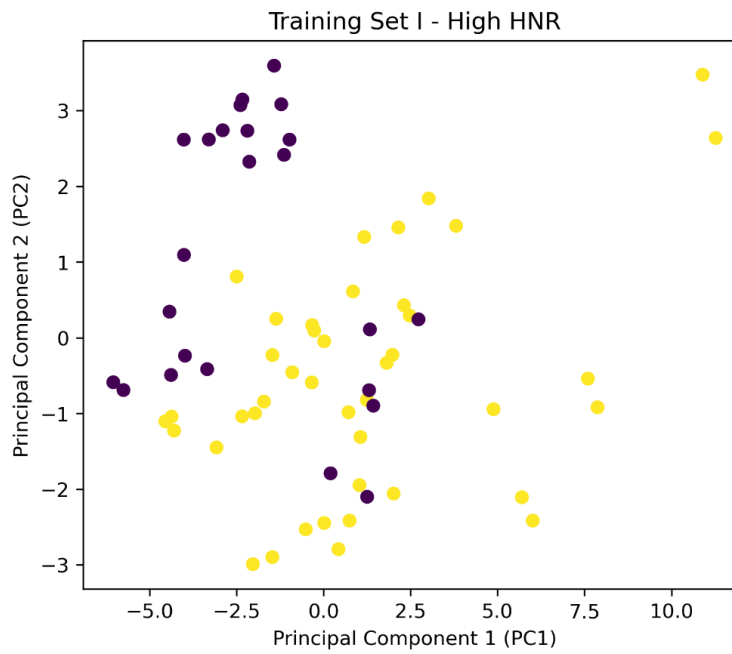


Figure 1: Training Set 1: High HNR

Figure 1 Description:

The above figure shows a scatter plot of the first two principal components (PC1 and PC2) obtained after applying PCA on the normalized High-HNR training dataset. Each point represents a data sample, and different colors show different class labels.

The data is spread widely along PC1, indicating that PC1 captures a large portion of the variance in the dataset. PC2 captures additional variation, but the spread along this axis is relatively smaller compared to PC1. The two classes (represented by different colors) show partial separation: One class is more concentrated in the upper-left region (negative PC1, higher PC2 values). The other class is more dispersed across the plot, especially along positive PC1 values.

There is some overlap between the classes, particularly around the center region, suggesting that the first two principal components alone do not perfectly separate the classes.

Figure 1 Findings:

PCA has managed to downsize the dataset to two dimensions without unimportant variance. The most influential component is PC1 that makes the highest contribution (data variability). Though it can be said that there is some visible clustering, the overlap represents that there are some factors. To begin with, additional principal components can be required to have a higher separation. Secondly, this feature space is not linearly separable into the classes.

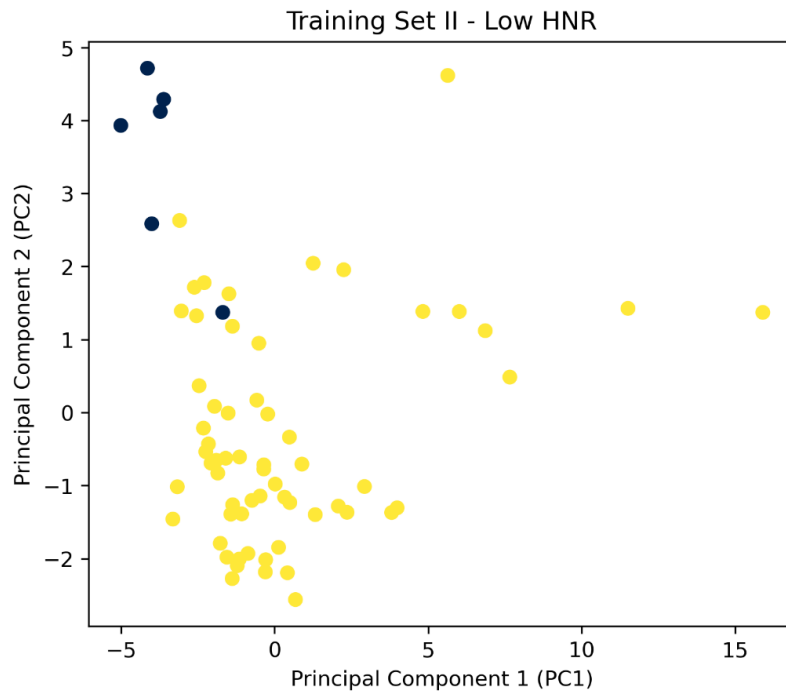


Figure 2: Training Set 2 Low HNR

Figure 2 Description:

The figure is a scatter plot of the first principal component (PC1) versus the second principal component (PC2) of the Low-HNR training data. Moreover, the data points are also broadly distributed on PC1 suggesting that PC1 measures the greatest percentage of variance in the data. There are some points that seem distant to the central cluster especially on the positive side of PC1, which shows the existence of outliers or extreme values. PC2 presents a relatively smaller degree of variation since the majority of the points become more condensed in the vertical direction than in the horizontal direction.

Figure 2 Findings:

The largest component is PC1 that describes the majority of changes in the data. Besides, the separation between classes is present to some extent, especially in the upper-left part of the image, yet the classes cannot be entirely separated. Having the overlap regions shows that the initial two major components are not ideal on their own to be used to classify.

2. Task 2: SVM Classification on High-HNR Group

(a) Data Preparation

The results indicate that the mean of each attribute in the normalized training data equals a value of about zero and standard deviation equals one, which proves that the StandardScaler was properly fitted on the training data. In the case of the validation set, the standard deviation and the mean are not exactly equal to zero and one, yet they are close to it, which is reasonable, as the validation data

was scaled with the help of the scaler trained on the training set instead of being utilized to fit it. This all means that there was a correctly applied normalization and no data leakage was observed.

(b) Model Performance on different combination of parameters

I have test the model on three different combination of parameters and the results are given below:

C	γ (Gamma)	Accuracy
1	1	0.7857
20	0.5	0.7857
0.5	0.1	1.0000

(c) Selection of combination of parameter values

I have selected the (0.5, 0.1) combination of parameter values because they produce nearly perfect accuracy of 100%. There may also be a possibility of overfitting the data.

(d) Model Performance on test data

Confusion matrix: $\begin{bmatrix} 0 & 12 \\ 0 & 19 \end{bmatrix}$

Top left -> True Negative: The model failed to identify any instance of class 0.

Top right -> False Positive: There were 12 samples that were actually Class 0, but the model incorrectly labeled all of them as Class 1.

Bottom left -> False Negative: The model did not miss any Class 1 samples (it never accidentally called them Class 0).

Bottom right -> True Positive: The model correctly identified all 19 instances of Class 1.

Having tested the model with the selected parameter values, I have achieved 0.61% of accuracy. The confusion matrix reveals that the model just become a majority-class classifier, predicting Class 1 for every sample and failing to identify any instances of Class 0. Despite an accuracy of 61%, the model lacks actual predictive power and suffers from severe overfitting, as evidenced by the sharp drop from perfect validation performance of 100% to this "all-one" phenomenon on the test set. This suggests the model has memorized training noise or specific feature distributions that do not generalize to unseen data.

3. Task 3: SVM Classification on Low-HNR Group

(a) Data Preparation

Using a 20% split ratio, the dataset was divided into a training set (II) and a validation set. To prevent data leakage, a StandardScaler was fitted exclusively on the training set, which successfully achieved a mean of approximately 0 and a standard deviation of 1.0 across all features. In comparison, the validation set exhibits varying means -0.54 to 0.67 and standard deviations 0.44 to 2.29, as it was transformed using the training parameters rather than its own; this confirms that the validation data was correctly treated as unseen during the normalization process.

(b) Model Performance on different combination of parameters

The following table summarizes all accuracies across different combination of parameter values. Interestingly, all accuracies are same despite having different set of parameter values.

C	γ (Gamma)	Accuracy
1	1	0.8571
20	0.5	0.8571
0.5	0.1	0.8571

(c) Selection of combination of parameter values

Since all three parameter combinations produced an identical accuracy of 0.8571, it is difficult to make a selection solely based on performance. In this case, I believe when accuracies are tied while using SVM, we can simply choose the simplest combination of parameter values that promote best generalization and avoid overfitting.

(d) Model Performance on test data

Upon applying the optimized SVM model to the independent test dataset, the model achieved an accuracy rate of approximately 80.65%. The discrepancy between the 80.65% accuracy and the confusion matrix highlights a majority-class bias. Because the test set is imbalanced (consisting of 25 Parkinson's cases and only 6 healthy cases), the model has reached a high accuracy score simply by predicting "Class 1" for every single input. This indicates that while the model is highly effective at capturing the majority class, it has zero sensitivity toward Class 0. In a clinical context, such as Parkinson's detection, this would be considered a "failed" model because it cannot distinguish between a healthy individual and a patient; it effectively treats every subject as having the condition.

4. Task 4: Cross Model Check

(a) Evaluate the HighHNR Group Model

Accuracy: 0.9032258064516129

Confusion matrix:

```
[[ 3  3]
 [ 0 25]]
```

Testing the High-HNR-trained model on the Low-HNR group test set yielded an improved accuracy rate of approximately 90.32%.

In case of Class 0 (Healthy), 6 real healthy samples were considered in the evaluation of the model. Among them, 3 samples were appropriately identified as healthy which are known as True Negatives (TN). But the rest of the 3 samples had been wrongly identified as Parkinson and this is termed as False Positives (FP). This translates to 50% classification accuracy of the healthy group, which means that this model fails to classify the healthy people with high reliability. Even with this limitation, it indicates an improvement of the above results when there were no healthy samples that were properly identified. In the case of Class 1 (Parkinson's), the model was supported by 25 real Parkinson samples. Every 25 samples were rightly classified as Parkinson, which is included as True Positives (TP). This class did not have any misclassification as the number of False Negatives (FN) was zero. This gives the Parkinson class a sensitivity of 100% (recall), indicating that the model is quite effective in the process of identifying people with the disease.

The 90.32 accuracy jump is a result of the ability of the model to accurately isolate half of the healthy individuals. Although this model continues to bias towards the majority class (Philadelphia reference to the 3 False Positives), the fact that the features learned on the High-HNR training data transfer some cross-group generalization to the Low-HNR data indicates that the model has some degree of cross-group generalization.

(b) Evaluate the Low-HNR Group Model

Accuracy: 0.6129032258064516

Confusion Matrix:

```
[[ 0 12]
 [ 0 19]]
```

The SVM model that was trained with Low-HNR group was tested with the normalized High-HNR test set. The model had a moderate performance of 61.29%.

The confusion matrix indicates that 12 healthy samples (Class 0) were mistakenly classified to be Parkinson (False Positives), and zero True Negatives, indicating that the model entirely failed to distinguish between healthy people. Conversely, all 19 cases of Parkinson (Class 1) were correctly identified (True Positives), and no cases were False Negatives, which gives 100% sensitivities to identify cases of Parkinson. Such findings suggest that the model is very skewed in predicting the Parkinson class and fails to predict well when used on another group (High-HNR). This is an indication of weakness and brings to attention the effects of data differences on the performance of models.

(c) (i):

The StandardScaler fitted on the test set was used to normalize the set. In particular, the High-HNR test set was de-normalized with the mean and standard deviation calculated with the High-HNR training set and the Low-HNR test set normalized with the parameters calculated with the Low-HNR training set. This guarantees that the test data is scaled analogous to that of the training data and eliminates leaking of data.

The SVM training model that had been trained on the Low-HNR training set was tested on the normalized High-HNR test set. To normalize, High-HNR test data was scaled on the basis of the mean and standard deviation which was computed on the High-HNR training set so that consistency and leakage of data is avoided. To evaluate the generalization capability of the Low-HNR-trained model to other data distributions, this normalized High-HNR test set was then tested with the Low-HNR-trained model.

c) (ii):

In task 4 (a): I used an SVM model trained on the High-HNR training set (I). The reason for using this model resulted from its perfect performance on the training data (100%).

In task 4 (b): I used an SVM model trained on the Low-HNR training set (I) because fitting the model on all three combination of parameter values produced same results. Therefore, I selected this model for having same performance across different parameter values.

(d) Conclusion

(i) Comparison of Cross Model Performance

In the first case, the model accurately categorized 3 out of 6 healthy samples (50%), and all 25 Parkinson samples (100%), with a more balanced classification between the 2 classes. On the opposite

hand, in the latter case the model did not get any of the healthy samples (0%), but on the other hand it did get all of the 12 Parkinson samples, and all the Parkinson samples were true.

Despite the fact that both the models are 100% sensitive in the Parkinson's class, their performance is very different in the case of the healthy class. The second model is obviously more biased to majority class which overblows the accuracy but conceals poor performance of the minority classes. This brings out the balance of classes that may cause the accuracy to act in the wrong way. The confusion matrices give a solid piece of evidence, indicating that the second model totally disregards healthy class but the first does not lose all its potential to differentiate. Thus, the former model is more successful on the data of the other group, because it is more accurate and shows a higher level of balance among the classes.

(ii) Feature Difference and Generalization Issues

The performance difference indicates that there are variations in the feature distributions and patterns in the High-HNR and Low-HNR groups that impact generalization. PCA plots and previous analysis demonstrate that the datasets do not have equal spreads and clusters. For example, the correlations between features do not hold across the groups. Such differences may render it challenging to transfer patterns learnt on one group to the other with a model trained on a different group. Following are some of the noted factors that influence performance. To begin with, certain characteristics can vary or even have varying statistics in the two groups. Secondly, class segregation might be different in both datasets. Finally, the model can also acquire group-specific decision limits which are inapplicable.

That is why one model is more generalized compared to the other. The former model trained on stronger and generalizable patterns, whereas the latter model is overfitting to its own population and is not capable of adjusting to the new information, particularly on the minority population.

In a nutshell, the former model has a superior cross-group generalization whereas the latter is highly biased against the majority group. The findings point to the influence of the class imbalance and the dissimilarity between datasets that restrict the capacity of models to generalize to other groups.